



Physics and Chemistry • 1st year of Bachillerato • The atom and the periodic table • Class 2 of 6 • With April

The colour of flames



Class video: <https://youtu.be/CsWFn7bYBbU>

The code leads to the video.

Name: _____ Date: _____

Before watching

1. Lithium, sodium and potassium belong to the same family and are very alike. If a little of the salt of each one is put in a flame, will they give the same colour? Place your bet and explain why you think so.
2. When a salt colours a flame, what do you think sets the colour: the metal, the chlorine in the salt or the fire itself? How could you check?

While you watch the video

1. What did lithium, sodium and potassium have in common?

2. In the table below, write down the colour each metal gives the flame.
3. Why is the test done with the burner's blue flame and not the yellow one? What is that yellow?
4. Why are wooden splints used, a different one for each salt, rather than a metal wire?
5. Write what each pictogram warns about: the skull, corrosion, the exclamation mark, and the fish and tree.
6. Why is this test not done at home? What went wrong in the methanol accidents?
7. **Before Abril tells you:** what if the colour came from the chlorine? Design a test to find out. Say what you would change and what you would keep the same.
8. Why does the idea that the colour is "down to the fire" not hold up?
9. An unknown salt gives a deep red flame. It could be lithium or strontium. How would you find out?
10. What is the point of testing a splint dipped only in distilled water? And why distilled?

Lithium	Sodium	Potassium	Calcium	Strontium	Barium	Copper

In the lab: the unknown salt

Goggles on at all times, hair tied back, blue flame, a new splint for each solution and, when you finish, every splint into the beaker of water.

1. What we keep fixed. Before lighting the burner, write down what you will keep the same in every test.

What we keep fixed	How we manage it

2. What we see. The control goes first.

Solution	Colour	Intensity (faint, medium or strong)	Does it last or fade at once?	Notes
Distilled water (control)				
Lithium chloride				
Sodium chloride				
Potassium chloride				
Calcium chloride				
Strontium chloride				
Copper chloride				
Sodium sulfate				
Sodium carbonate				
Unknown X				

3. After testing.

1. What colour does the wood alone give? Can you notice it in any of the other flames?
2. Sodium with chloride, with sulfate and with carbonate: does the colour change? And the intensity? What does that tell you about the chlorine?
3. The sodium sulfate and carbonate were at 0.25 mol/L, and the chloride at 0.5. Why do you think they are prepared like that?
4. What is X? How sure are you (sure, fairly sure, not very)? Why?
5. What would you need to be completely sure?
6. Was there anything you could not keep fixed? Could it have changed the result?

Practice

1. The four flames A, B, C and D at the end of the video come from salts of sodium, lithium, copper and potassium, shuffled. Say which is which and what you look at.
2. Design an experiment to find out whether the red of a lithium salt comes from the lithium or from the other part of the salt. Say what you change, what you keep the same, what control you use and what result you would expect if the red came from the lithium and if it came from the other part.

3. Below potassium, in the same family, are rubidium and caesium. Try to predict the colour of their flame from their place in the table, and say whether it can be done.

A risk-free mission

Next time you watch fireworks, as a spectator and from a distance, look at their colours and write down which metal you think makes each one.

Colour you see	Metal you think makes it

Keep thinking

Today we found out what sets the colour by changing one thing at a time. But some things cannot be changed at will: the past, a distant star, a person's life. How do we find out what causes what when we cannot do the experiment?